Quadratic AI Supply Chain Exercises

📌 Objective

This file documents analysis performed using **Quadratic AI** on supply chain datasets. The exercises simulate real-world challenges such as incomplete deliveries, customer reliability, and bottleneck identification, showcasing how AI-driven analytics can generate actionable business insights.

🛠 Analyses Conducted

1. **Revenue Loss from Undelivered Orders**
   - Quantified financial impact of incomplete deliveries.
   - Highlighted risk areas for executive decision-making.
2. **Customer OTIF (On-Time, In-Full) Discrepancies**
   - Identified customers with the largest reliability gaps.
   - Prioritized accounts requiring intervention to maintain retention.
3. **Low 'In-Full' Product Categories**
   - Pinpointed categories consistently underperforming in fulfillment.
   - Suggested deeper root-cause analysis (supplier reliability, inventory).
4. **Average Delay Time**
   - Measured mean delay across late deliveries.
   - Provided benchmarks for logistics teams to improve scheduling.
5. **Supply Chain Bottlenecks**
   - Linked low 'In-Full' rates to warehouse capacity and supplier lead times.
   - Recommended targeted process improvements.

📊 Stakeholder Value

- **Executives:** Visibility into revenue leakage and customer risk.
- **Planners:** KPIs to prioritize corrective measures.
- **Warehouse Teams:** Delay metrics to optimize operations.
- **Customers:** Improved reliability and service levels.

🚀 Business Impact

- Reduced reporting cycles with automated KPI generation.

- Improved customer retention through proactive risk identification.
- Strengthened negotiation power with reliable delivery metrics.